

1. RESEARCH OBJECTIVE

Evaluate how state-level tax policy interventions affect cigarette consumption patterns across different U.S. regions over a 30-year period, examining the relationship between taxation policies and smoking behavior.

1.1. PROJECT SCOPE

Dataset:

- 50 U.S. States × 30 Years (1,500+ observations)
- Multi-dimensional panel data including:
 - Per-capita cigarette consumption
 - State tax as % of retail price
 - Federal tax as % of retail price
 - Demographic and socio-economic variables
 - Average cost per pack

Policy Context:

- 1998 Master Settlement Agreement (MSA)
- 2009 Federal Tax Increase
- 2019 Tobacco 21 Law
- Regional policy variations across states

1.2. DATA PREPROCESSING & FEATURE ENGINEERING

- Cleaned and validated multi-year panel datasets across all 50 U.S. states.
- Engineered tax percentage, regional aggregation, and temporal trend features for downstream analysis.
- Standardized state-level variables and prepared structured datasets for clustering, geospatial analysis, and visualization workflows.

1.3. ANALYTICAL METHODS

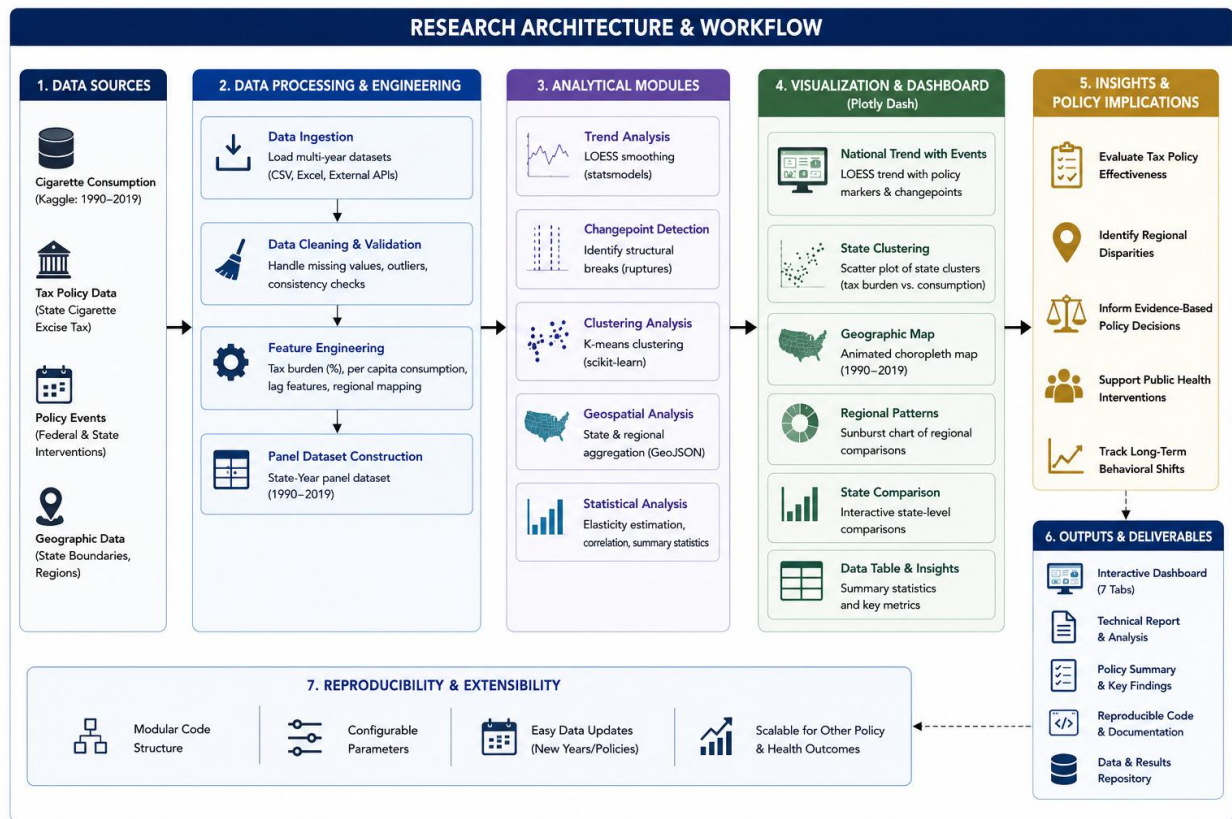
Statistical Techniques:

- LOESS Trend Analysis with policy event markers
- Changepoint Detection (ruptures algorithm) to identify structural breaks
- K-Means Clustering to group states by tax burden and consumption patterns
- Correlation Analysis across economic, demographic, and policy variables
- Elasticity Analysis examining tax-consumption relationships
- Geospatial Visualization showing regional disparities

Technical Implementation:

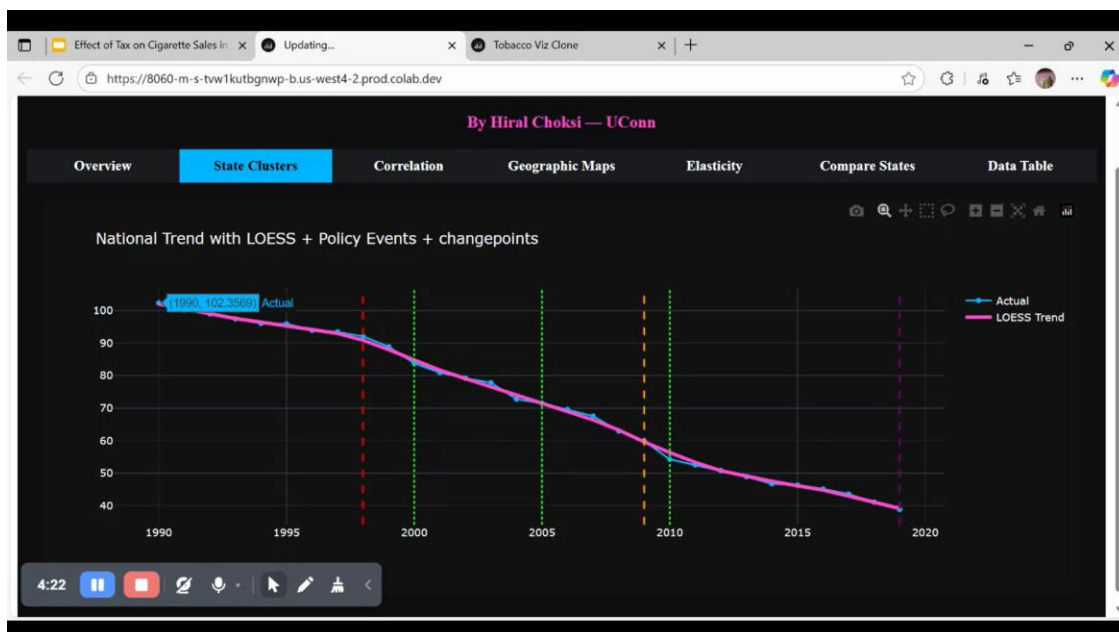
- Interactive 7-tab Dash dashboard
- Animated choropleth maps (GeoJSON integration)
- State-level comparison tools
- Dual-theme interface (accessibility)

1.4. RESEARCH ARCHITECTURE & WORKFLOW



1.5. KEY VISUALIZATIONS

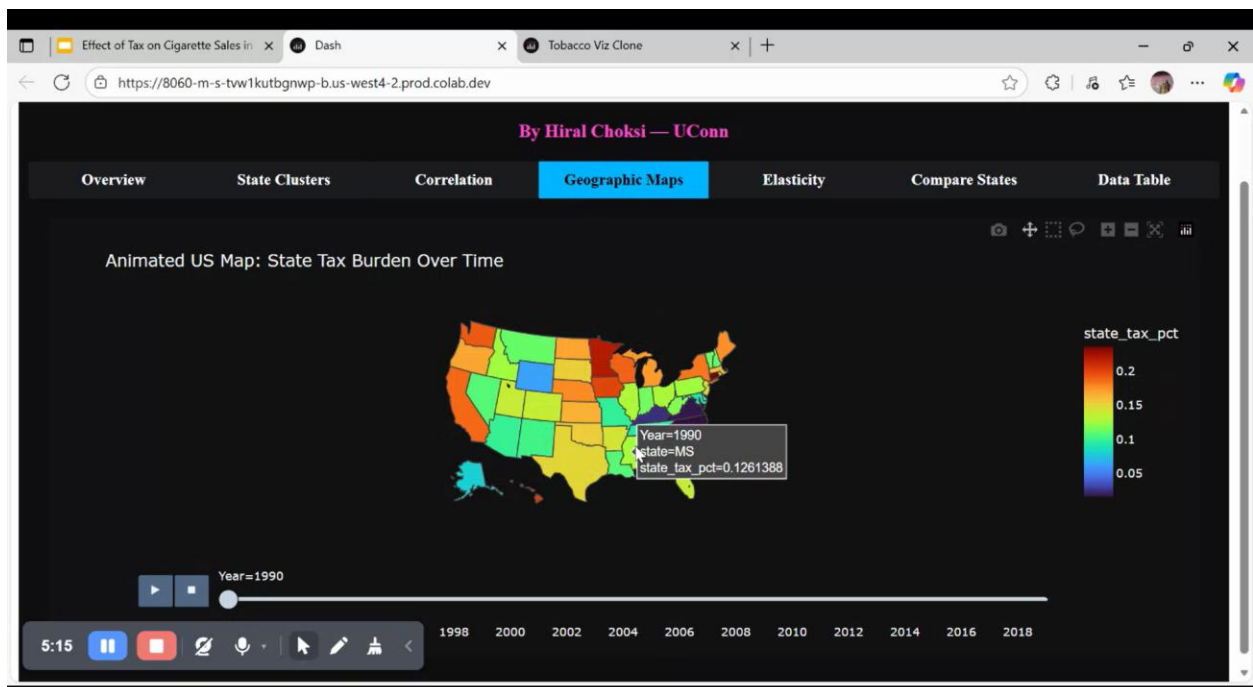
National Trend with Policy Interventions

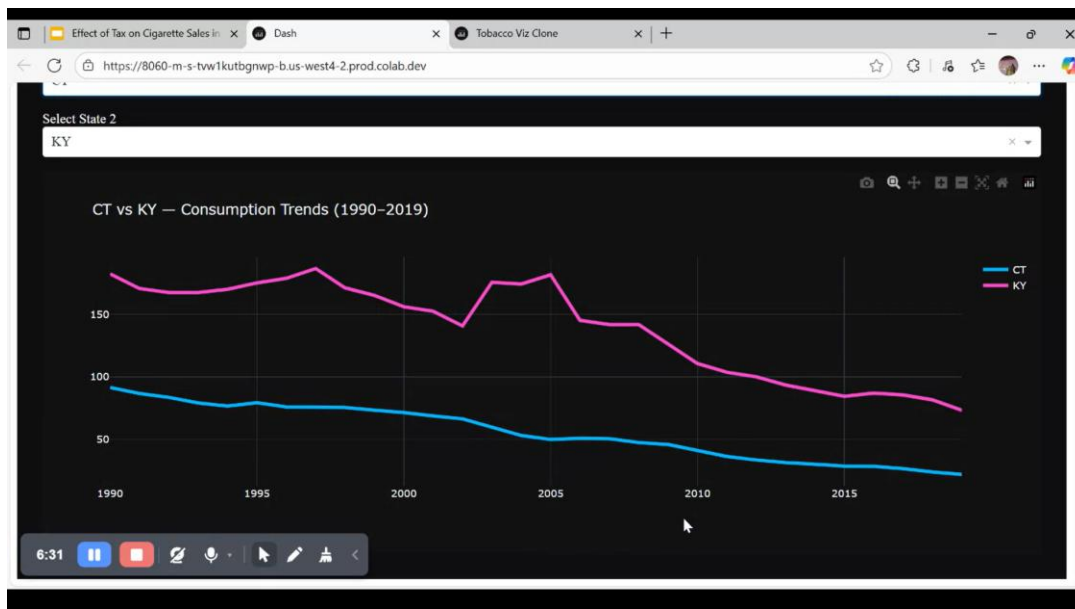


State Clustering Analysis

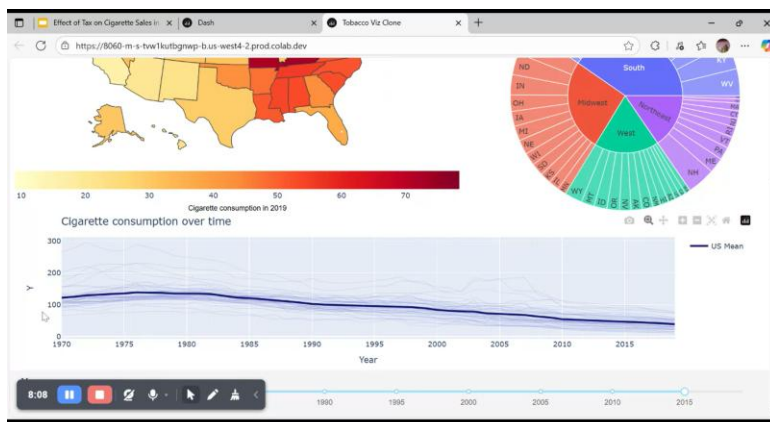
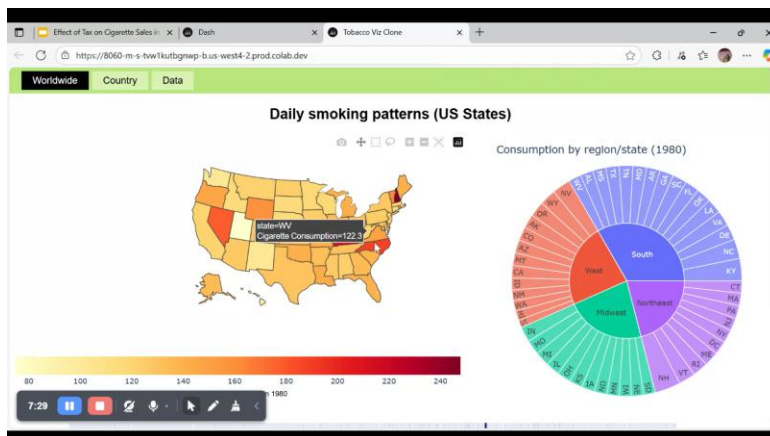


Geographic Distribution





Regional Comparisons



2. KEY FINDINGS

- National cigarette consumption shows a clear long-term decline with observable structural shifts near major policy intervention periods.
- States cluster into 4 distinct groups based on tax policy and consumption behavior, revealing regional policy approaches
- Observed negative relationship between tax burden and cigarette consumption trends across states.
- Regional disparities persist: Southern states maintain higher consumption despite national trends
- Temporal and spatial patterns reveal both immediate policy impacts and longer-term behavioral shifts

2.1 RESEARCH RELEVANCE TO POLICY EVALUATION

This project demonstrates:

- **Panel data analysis** techniques applicable to evaluating interventions across geographic units over time
- **Policy impact assessment** methodology using temporal markers and changepoint detection
- **Geospatial analysis** to identify regional patterns and disparities in policy outcomes
- **Statistical rigor** in examining intervention-outcome relationships
- **Interactive visualization** for communicating complex policy research findings

3. TECHNICAL STACK

- **Languages & Libraries:**
Python • Pandas • NumPy • Plotly • Dash
- **Statistical Methods:**
statsmodels (LOESS) • ruptures (changepoint) • scikit-learn (K-means, clustering)
- **Geospatial:**
GeoJSON • Choropleth mapping • Regional aggregation
- **Data Engineering:**
Multi-year panel data processing • Feature engineering (tax percentages, regional mapping) • Data validation and cleaning